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Grid harmonic impact of multiple electric vehicle fast charging

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ABSTRACT

Fast charging is perceived by users as a preferred method for extending the average daily mobility of electric vehicles (EV). The rated power of fast chargers, their expected operation during peak hours, and clustering in designated stations, raise significant concerns. On one hand it raises concerns about standard requirements for power quality, especially harmonic distortion due to the use of power electronics connecting to high loads, typically ranging from 18 to 24 kW h. On the other hand, infrastructure dimensioning and design limitations for those investing in such facilities need to be considered. Four sets of measurements were performed during the complete charging cycle of an EV, and individual harmonic's amplitude and phase angles behaviour were analysed. In addition, the voltage and current total harmonic distortion (THD) and Total Demand Distortion (TDD) were calculated and the results compared with the IEEE519, IEC 61000/EN50160 standards. Additionally, two vehicles being fast charged while connected to the same feeder were simulated and an analysis was carried out on how the harmonic phase angles would relate. The study concluded that the use of TDD was a better indicator than THD, since the former uses the maximum current (I_L) and the latter uses the fundamental current, sometimes misleading conclusions, hence it is suggested it should be included in IEC/EN standard updates. Voltage THD and TDD for the charger analysed, were within the standard's limits of 1.2% and 12% respectively, however individual harmonics (11th and 13th) failed to comply with the 5.5% limit in IEEE 519 (5% and 3% respectively in IEC61000). Phase angles tended to have preferential range differences from the fundamental wave. It was found that the average difference between the same harmonic order phase angles was lower than 90°, meaning that when more than one vehicle is connected to the same feeder the amplitudes will add. Since the limits are dependable on the upstream short circuit current (I_{SC}), if the number of vehicles increases (i.e. I_L), the standard limits will decrease and eventually be exceeded. The harmonic limitation is hence the primary binding condition, certainly before the power limitation. The initial limit to the number of chargers is not the power capacity of the upstream power circuit but the harmonic limits for electricity pollution.

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1. Introduction

The paradigm change from centralised unidirectional electricity flow from plants to consumers into a distributed bidirectional electricity flow system, poses new challenges. These include the need for new operational strategies, models or simulation tools, and better infrastructure design (technological development and adaptation to a wider distributed grid). The interaction between distributed agents including smart houses or electric vehicles (EV), and the electricity grid has prompted the interest of researchers, industry and policy makers. These interactions between technical agents are supported by social/economic agents like prosumers, retailers, Distributed System Operators (DSO) or service companies, which are crucial to drive the interactions at a technical level. Standardisation requirements and a better understanding of interoperability phenomena, especially the impacts that large scale integration of distributed agents may have, are subjects of interest for industry, utilities, regulatory and policy making organisations. Among others, interoperability studies focus on enabling universal connectivity, batteries and EV interaction with the grid and other agents, and identifying gaps in standards or technologies.

Abbreviations: A_x, X axes component of vector A; A_y, Y axes component of vector A; B_x, X axes component of vector B; B_y, Y axes component of vector B; EV, electric vehicle; $E(\theta)$, phase angle mean value; I_1, fundamental current; I_h, individual current harmonic order; I_L, maximum demand load current at PCC; I_{ref}, reference current; I_{SC}, maximum short-circuit current at PCC; J, load; PCC, point of common coupling; PQ, power quality; PWHC, partial weighted harmonic current; R, resulting amplitude of a vector; R_X, resulting amplitude of vector X axes component; R_Y, resulting amplitude of vector Y axes component; RSCE, short-circuit ratio; SOC, state of charge; TDD, total demand distortion; THD, total harmonic distortion; THD_I, current total harmonic distortion; THD_V, voltage total harmonic distortion; Z, impedance; $U_{h\Sigma}$, resultant vector of a harmonic order; θ_h, phase angle of a vector related to the fundamental; $\sigma(\theta)$, phase angle standard deviation.

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The electricity industry has recommended Mode 3 EV charging (quick, single-phase or three-phase options) (IEC 61851) [1] as the preferred solution for all types of locations, making fast charging an important area to be addressed. Topics for the analysis by which EV charging impacts distribution networks can be listed as voltage regulation, harmonic distortion levels, unbalances, additional losses and transformers loss of lifetime. Battery chargers for EVs employ nonlinear switching devices which may result in significant harmonic voltage and currents injected into the distribution system. Literature reports different findings regarding the impact on power quality from EVs. Some authors state that distribution infrastructure can have limitations with EV charging supply even for relatively low EV penetration levels [2–5]. Other studies [6–11], suggest that low EV penetration levels, with normal charging rates, will have acceptably low harmonic levels and voltage variations, however fast charging rates could cause significant voltage harmonics and losses. Most of the studies tend to focus only on current harmonics, addressing as the main concern residential and normal chargers, as they are expected to have higher penetration. There is however, a very limited number of studies which analyse both voltage and current harmonics focusing on fast charging specifically performed in a cluster of chargers connected to the same feeder [12,13]. The motivation of the studies tends to be the same, high expected impact on energy demand and expected usage during peak hours.

Considerable literature focuses on the distribution networks especially residential networks [14,15], where EV charging could bring an additional severe power electronic load and associated power quality issues. The European standard for public power supply EN 50160 [16], sets conditions for: voltage magnitude variation, voltage harmonics, inter-harmonic voltage and voltage unbalance among others. All loads that are connected to the power network must provide a sufficiently low effect on the network, so that it does not cause a violation of the power supply conditions stated in this standard. This means that the EV chargers, once connected to a public network, must not influence the network operation to the extent that can cause deviation from the standard. In general, EV chargers have to fulfil requirements for loads that can be connected to an electric power network as described by the electromagnetic compatibility IEC 61000 series standards. These standards set the emission levels, including the harmonic currents or power factor that a charger is allowed to have. The standards applied to the low-power EV chargers are IEC 61000-3-2 [17] and IEC 61000-3-4 [18], which set limits to the harmonic emissions generated by the charger. Another study based on practical measurements of charging commercial EVs [8] presents a maximum THD_I of 17.3% for a level III charger at the end of the charge, and maximum of 19.2% for Level I and II also at the end. This publication acknowledges that TDD use would improve the conclusions regarding the distortion impact. Results from a case study in Portugal [9] found a THD_I of 11.6%, during the constant charging stage in a fast charging station integrated in a commercial facility.

A typical distribution network has a large number of different non-linear loads connected to it. Authors in Ref. [19] argue that adding EV chargers from different manufacturers may result in a variety of different harmonic patterns. The diversity of the patterns may lead to notable harmonic cancellation. This effect occurs when harmonics with different phase angles provide a sum in the magnitude that is smaller than the individual harmonics magnitudes. It is however rather complicated to evaluate this effect. Authors in Ref. [10] studying low voltage nonlinear loads also suggest that cancellation is more probable as the number of consumers and appliances increase. It has also been indicated that harmonic cancellation is more expected at higher harmonic orders, which can then account for the relatively minor THD_I decrease. In most papers, it is rather common that only the harmonics amplitude levels are observed, as the utilities are required to keep the harmonics levels under a given limit. Authors in Ref. [20] however note that if the diversity of chargers is not taken into account, the harmonic problems could be overestimated. One of the first papers in this area was actually presented by Ref. [21] where multiple different EV chargers in the network have been observed. It is reported, that 10% smaller harmonic current magnitudes were observed compared to the simple summing of magnitudes. Another study [22] applies a methodology which accounts for diversity of SOC and initial charging moments in California. The results indicate that accounting for variation in start-time and SOC in the analysis leads to reduced estimates of harmonic current injection. Authors argue that traditional methods do not account for these variations. Researchers show that from the point of view of the substation transformer, the impact of EV's is mainly one of power and energy, rather than harmonics. Analysis with real and imaginary components for each harmonic has been described in Ref. [7] which analyses 20 kW h charges and reports a THD_I over 40% at the connection point. The 11 kV medium-voltage network was simulated with 36 chargers, each at power level of 8.2 kV A, which makes it difficult to observe the total cancellation effect.

There is still a lack of studies focusing on clustering fast chargers and the impacts on both THD_I and THD_V (voltage total harmonic distortion) referring specifically to fast charging. It is critical to study the phase angles in order to understand how the amplitude of the harmonics measured will sum when considered part of a cluster (connected to the same feeder). The main goal of this study is to clarify the following questions: (i) investigate the voltage and current Total Harmonic Distortion (THD) caused by fast charging one single electric vehicle and standard limits compliance. (ii) how does the THD caused by fast charger/EV load vary along the charging cycle if at all?, and (iii) does the THD and TDD caused by charging two EVs simultaneously with the same fast charger decrease due to phase cancellation?

2. Theoretical background

Harmonic distortion is a deviation of the current or voltage waveform from a perfect sinusoidal shape. In the case of nonlinear loads, such as EV charge controllers, current distortion is very common due to the need of power electronics switches to convert power from AC to DC form. Introduction of these currents into the distribution system can distort the utility supply voltage and overload expensive electrical distribution equipment. In order to prevent harmonics from negatively affecting the utility supply, standards such as IEC 61000-3-12 [23]/2-4 [24] or IEEE Standard 519-1992 [25], were established with the goal of developing recommended practices and requirements for harmonic control in electrical power systems. These standards, widely adopted by the industrial and research communities, describe the problems that unmitigated harmonic current distortion may cause within electrical systems as well as the degree to which harmonics can be tolerated by a given system. Utilities are obliged to provide power quality whose limits among others depend on the level of voltage connection. End users on the other hand, are responsible for not degrading the voltage of the utility by drawing significant nonlinear or distorted currents. Power quality, specifically harmonic impact in Points of Common Coupling (PCC) is a subject of interest to both parties. The PCC with the consumer/utility interface is the closest point on the utility side of the customer's service where another utility customer is or could be supplied. The goal of applying the harmonic limits specified in the standards is to prevent one customer from causing harmonic distortions to another customer or the utility.

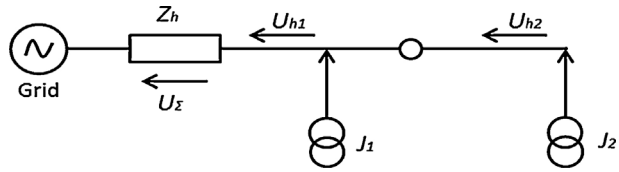

Fig. 1. Simplification example of two loads connection to the same grid feeder.

2.1. Harmonics

Harmonics topic theory is a well-covered subject. In practical terms however, it is difficult to assess the phase angles from each harmonic order, and therefore to make valid assumptions regarding the way they add up when multiple devices interact. Typically, probabilistic approaches are made [20,26], and often studies will treat the vector summation considering high uncertainty or the worst case scenario [27]. For this reason literature results and conclusions often differ. For example in Fig. 1, two loads J_1, J_2 connected to a grid with the impedance Z_h are considered.

The vectors U in Fig. 1, with harmonic voltage order h, will sum ($U_{h\Sigma}$) according to Eq. (1), where θ is the phase angle related to the fundamental.

$$U_{h\Sigma} = \sqrt{U_{h1}^2 + U_{h2}^2 + 2U_{h1}U_{h2}\ \cos\left(\theta_{h2} - \theta_{h1}\right)} \tag{1}$$

However if the angles are unknown and if no probability function exists for θ, one can use the properties of a uniform distribution to deduce the probability of a conservative summation by upper and lower deviation phase angles establishing as shown in Eq. (2), the limit between 0 and π where,

$$f\left(\theta\right) = \frac{1}{\pi}, \qquad \theta \in [0, \pi] \tag{2}$$

The following equations shows the expected mean value obtained by

$$E\left(\theta\right) = \int_0^{\pi} \theta f\left(\theta\right) \mathrm{d}\theta = \frac{\pi}{2} \tag{3}$$

and the corresponding standard deviation in the following equations is

$$\sigma\left(\theta\right) = \sqrt{\int_0^{\pi} \theta^2 f\left(\theta\right) \mathrm{d}\theta - \left[E\left(\theta\right)\right]^2} = \frac{\pi}{2\sqrt{3}} \tag{4}$$

This can give the upper and lower phase angles estimations as follows in the following equations:

$$\theta_{\text{upper}} = E\left(\theta\right) + \sigma\left(\theta\right) = \frac{\pi}{2}\left(1 + \frac{1}{\sqrt{3}}\right) \rightarrow 141.96^o, \qquad p = 78.86\% \tag{5}$$

$$\theta_{\text{lower}} = E\left(\theta\right) + \sigma\left(\theta\right) = \frac{\pi}{2}\left(1 + \frac{1}{\sqrt{3}}\right) \rightarrow 38.04^o, \qquad p = 21.13\% \tag{6}$$

In the event the statistical distribution or exact angles are known, they will add together when their difference is below 90° (add perfectly if 0°) or subtract if below (and cancel each other if 180°). Moreover the resulting angle of each sum can be calculated analytically by decomposition of the X and Y components. Consider two vectors A and B in Eqs. (7)–(8) and (9)–(10), where A_1 and A_2 are the vector's amplitude and θ_1 and θ_2 are the corresponding angles:

$$A_x = A_1\ \cos\theta_1 \tag{7}$$

$$A_y = A_1\ \sin\theta_1 \tag{8}$$

$$B_x = A_2\ \cos\theta_2 \tag{9}$$

$$B_y = A_2\ \sin\theta_2 \tag{10}$$

After obtaining the R_X (by adding the X and Y components), the resulting amplitude (R) in Eq. (11) and angle (θ_R) in Eq. (12) are given by

$$R = \sqrt{R_X^2 + R_Y^2} \tag{11}$$

$$\theta_R = \tan^{-1}\left(\frac{R_Y}{R_X}\right) \tag{12}$$

2.2. Standard IEEE 519, IEC 61000 and EN 50160

IEEE 519-1992 and IEC 61000-3-12/2-4 are respectively the American and International standards which apply to the case under study. Both discuss the impacts that harmonic distortion can have on distribution assets, particularly transformers, power cables, capacitors, metering, relaying and switch gear. Both propose limits for voltage and current distortions and limits for individual frequencies. IEEE 519 presents the voltage limits, and still makes a clear distinction between THD_I and TDD concepts. EN 50160 presents the main voltage level parameters and their permissible deviation ranges at the customer's PCC in public low voltage and medium voltage electricity distribution systems. However the load current is not relevant to EN 50160. Regarding the actual current harmonic limits, the European standards are the same as within IEC, hence only the latter will be referred to from this point. Table 1 shows the Voltage THD limits for different voltage levels. Similarly Table 2 shows the limits for the TDD and individual harmonics according to each voltage level. Tables 3 and 4 show the same limits set in IEC 61000-3-12/2-4 respectively.

I_{SC}/I_L ratio shows the relative size of the load compared to the utility system. Power systems in a given point (under a linearity hypothesis) can be transformed into a Thevenin equivalent with the related impedance. The short circuit, which may also be expressed in short-circuit power (SCP) in MVA, at that point "quantifies" the equivalent impedance of the network. If it is high (i.e. low SCP)

Table 1
Voltage distortion limits set in IEEE 519-1992.

Bus voltage at PCC	Individual voltage distortion (%)	Total voltage distortion THD (%)
69 kV and below	3	5
69.001 kV through 161 kV	1.5	2.5
161.001 kV and above	1	1.5

Table 2
Maximum harmonic current distortion in percent of I_L set in IEEE 519-1992.

Individual harmonic order (odd harmonics)						
I_{SC}/I_L	<11	$11 \le h < 17$	$17 \le h < 23$	$23 \le h < 35$	$35 \le h$	TDD
<20	4.0	2.0	1.5	0.6	0.3	5.0
20 < 50	7.0	3.5	2.5	1.0	0.5	8.0
50 < 100	10.0	4.5	4.0	1.5	0.7	12.0
100 < 1000	12.0	5.5	5.0	2.0	1.0	15.0
>1000	15.0	7.0	6.0	2.5	1.4	20.0

I_L—Max. demand load current and I_{SC}—short circuit current. TDD—total demand distortion, harmonic current distortion in % of maximum demand load current. Even harmonics are limited to 25% of odd harmonic limits above.

Table 3
Maximum harmonic current distortion in percent of I_L set in IEC 61000-3-12.

Minimum RSCE	Admissible individual harmonic current I_h/I_{ref} (%)				Admissible harmonic parameters (%)	
	I_5	I_7	I_{11}	I_{13}	THC/I_{ref}	PWHC/I_{ref}
33	10.7	7.2	3.1	2	13	22
66	14	9	5	3	16	25
120	19	12	7	4	22	28
250	31	20	12	7	37	38
≥350	40	25	15	10	48	46

The relative values of even harmonics up to order 12 shall not exceed 16/*h*%. Even harmonics above order 12 are taken into account in THC and PWHC in the same way as odd order harmonics. Linear interpolation between successive R_{SCE} values is permitted. R_{SCE}—short-circuit ratio; I_h—harmonic current component; I_{ref}—reference current; THC—total harmonic current; PWHC—partial weighted harmonic current.

Table 4
Voltage distortion limits set in IEC 61000 2-4.

Harmonic order *n* (non multiples of 3)	Class 1 μ.n [%]	Class 2 μ.n [%]	Class 3 μ.n [%]
5; 7	3; 3	6; 5	8; 7
11; 13; 17	3; 3; 2	3.5; 3; 2	5; 4.5; 4
THD_V	5%	8%	10%

Class 1: compatibility level lower than public (laboratory instrumentation, some protection equipment, etc.). Class 2: compatibility level equal to public (any equipment designed for supply from public networks). Class 3: compatibility level higher than public (equipment in the presence of welding machines, rapidly varying loads, large converters, etc.).

the network is "weak" and the voltage is affected by the (harmonic) currents; if it is high (infinite), the impedance is zero and the network is strong and the voltage is not affected. It is therefore necessary to calculate or to measure the short circuit current (I_{SC}) at the PCC where the measurements are intended. TDD is very similar to THD, except for the denominator as shown in Eqs. (13) and (14). In TDD, harmonics are expressed as % of I_L (maximum demand load current) whereas THD express harmonic content as % of I_1 (fundamental current). For the I_L it is recommended to consider the maximum averaged current of at least a 15–30 min interval of the last 6 months for a given consumer.

$$THD_I = \frac{\sqrt{I_2^2 + I_3^2 + I_4^2 + I_5^2 + \cdots}}{I_1} \quad (13)$$

$$TDD = \frac{\sqrt{I_2^2 + I_3^2 + I_4^2 + I_5^2 + \cdots}}{I_L} \quad (14)$$

Usually the THD is measured first and then a comparison is made to the limits, if there is a problem, then the TDD is calculated. It is rarely needed to convert to the TDD and % of I_L, which is why the THD concept is much better known.

2.3. Test design

Four measurements were performed on a full electric vehicle using a fast charger. As a measurement device the Fluke 437 Series II [28], 400 Hz Power Quality and Energy Analyser was used, set with 0.25 s time step data acquisition. The harmonics were registered until 2500 Hz. The resolution and accuracy of the THD for both voltage and current is 0.1% and ±2.5% of the reading respectively, whereas for the phase angles it is 1° with an accuracy of $\pm n \times 1°$ (where n is the harmonic order). A commercial fast charger [29] was connected to a max 63 A outlet, 230 V, 50 Hz at one end and at the other, with the chargers manufacturer's cable, to a full electric vehicle VW model E-Up, with an 18.7 kW h battery pack. The vehicle was discharged over random driving cycles and the different measurements were performed on different days. The laboratory temperature was approximately 25 °C in all measurements. All loads inside the car were disconnected (air conditioning, radio, lights). The balance of the system was verified successfully. The 4 sets of measurements were performed from different low states of charge 8%, 7%, 5%, and 10% respectively, to 100% SOC in a maximum of 42 min.

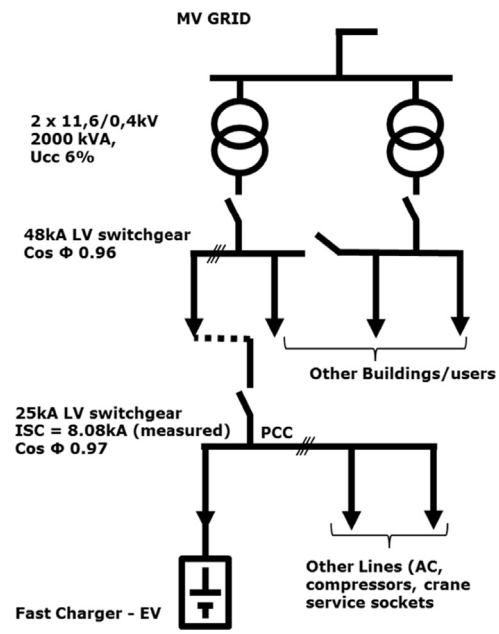

Fig. 2. Simplified single line diagram of upstream electricity grid, from PCC to MV level.

Before starting the measurements, in addition to phase sequence verification, an initial verification procedure was performed. This was intended to mitigate the fact that not all measurements were recorded at the same time, and that no voltage control source was used. A set of three files were recorded per measurement: (i) no load, (ii) only with the charger connected and (iii) with load. This was intended to verify: frequency fluctuation, voltage fluctuation, THD_V with no load and THD_V only with the fast charger connected.

The upstream grid representation is shown in Fig. 2, as well as the point (PCC) where the measurements were performed. The PCC in theory will often be at the medium voltage level, i.e. the primary of the distribution transformer serving the users, irrespective of transformer ownership or the location of the metering system. In practical terms however it is often more secure or accessible to perform such measurements on the transformer secondary, as is the case presented in this analysis. The present study considers that a distinction of consumers would be done at the PCC point shown in Fig. 2. An impedance meter (Impmeter 2) was used to record the I_{SC} at the PCC and with each I_L from each measurement the standard limits were identified.

To accomplish the first and second goals of this research, after the measurements were performed, PowerLog 4.2 software [28]

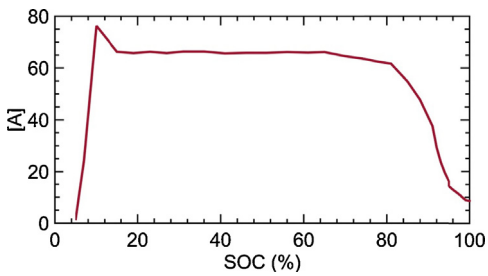

Fig. 3. Current behaviour during a charging cycle (L_1).

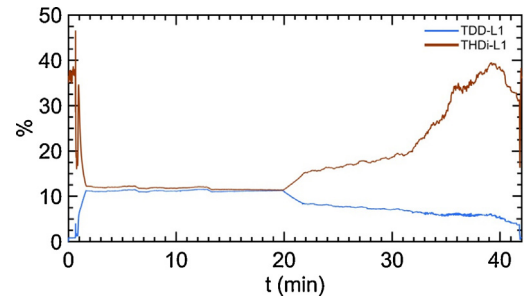

Fig. 4. THD_I and TDD during a complete charging cycle (L_1).

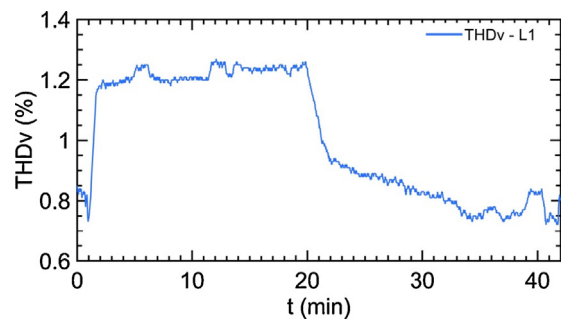

Fig. 5. THD_V during a complete charging cycle (L_1).

was used to import and verify the reading. The data from the four sets of measurements were then exported to spreadsheets. Values of THD were observed and compared with TDD, the ARMS current during charging cycle was registered and all shown in the result and discussion chapter. From the absolute and relative values, a comparison with the standard limits was possible. It was also important to obtain these values in order to simulate the THD and TDD with one and two electric vehicles working together. Since the system is balanced, the analysis was only performed in one phase. To pursue the third challenge of this study, apart from the amplitudes of the harmonics, also their phase angles were examined. Using the Crystal-ball excel add-in, the time series of each angle and phase were submitted to a curve fit calculator (based on Anderson–Darling statistical test). The phase angles from each frequency were analysed and their corresponding statistical distributions were analysed regarding their range differences. After this analysis, a simulation was carried out with the goal of obtaining the corresponding TDD resultant from charging one vehicle or two vehicles simultaneously. As inputs for the simulation, the complete data set of approximately 40 min was considered from two different load measurements from one single phase. Using Crystal-ball, both absolute values of amplitudes, and the phase angles statistical distributions were used to apply Eq. (1) to the harmonics. By using Eq. (8) the two TDD were found and compared.

3. Results and discussion

3.1. Results from measurements

Measurements were taken for approximately 40 min with 0.25 s time steps, generating over 9000 records for each unit. A simplification of 7 events during the charging cycle is given in Tables A1–A4 of the Annex. All the odd harmonics until the 25th are shown, as well as the THD_I of each phase in the corresponding minute. In all measurements it is possible verify a high predominance of the 3rd, 5th, 7th, 9th, 11th, 13th harmonics. Furthermore, higher variability in the phase angles is observable in the 3rd and 9th harmonic (Figs. A1–A6), whereas the other odd harmonics present a relatively narrow angle range. The THD_I tends to increase at the end of the charging cycle, which can be explained by the decrease of the current at the end of the cycle, as shown in Fig. 3. This however, can be misleading if only the THD_I is considered, since it considers the fundamental current as reference. The THD_I can reach as high as 40% in the 4th measurement shown in Table A6, however the current which it refers to is below 10 A instead of 67.5 A during most of the charging, where the TDH_I is approximately 11%–12%. For this reason, the analysis of the TDD which considers the maximum current instead of the fundamental is the most elucidative of that which is under study. From Figs. 3 and 4 three distinct stages can be observed: First during the first 2–3 min of charging with very high THD_I peaking to more than 45% while TDD starts low, followed by a second stage with a constant behaviour where TDD and THD present very close values of 11.5% to 12.5%. A third stage can be distinguished during the last 15 min corresponding to 77%–100% SOC where the current starts decreasing, making the THD reach a maximum of 41% and TDD drop to 0%. The graphic in Figs. 3 and 4 corresponds to just over 40 min of charge.

Values of both TDD and THD_I are coherent with others presented found in literature [8,9]. TDD calculations were based on the maximum current of 67.5 A demand during the charging cycle (even though a peak of 76.7 A was recorded, it only lasted 40 s and therefore was neglected). Maximum value of TDD was 13.12% and for the THD_I it was 47%. The readings enhance the need to separate the analysis using the fundamental and the maximum demand current. Regarding the THD_V, Fig. 5 presents the complete behaviour during the charging cycle for phase L_1.

The time evolution is coherent with the current one as shown in Fig. 3. The voltage starts decreasing at approximately 75% SOC, which also happens with the current, meaning that this stage has a lower power being fed into the battery (coherent with lithium-ion cells technology charging curve). The lower power in this stage is the reason why the last 12 min only charge about 15% of the charge. The Voltage THD in the initial stage reaches 1.26% distortion and it stabilises 1.16% during the constant stage, dropping to 0.7% at the final stage, which are all below the 5% limit [25]. The harmonic histogram for both voltage and current can be observed in Fig. 6. Even harmonics were excluded due to their low contribution and only harmonics until the 25th are shown, since above this their values are negligible. One can verify that the odd harmonics are the most significant especially the 3rd, 5th, 7th, 9th, 11th and 13th. Their lower or higher relative values as shown in the picture do not reveal the total importance of their individual analysis, since their limits as set by the standards become lower as the frequency increases. For this reason all odd harmonics until the 25th were analysed. For the analysis of the phase angle range, each odd harmonic frequency was compared with the other measurements by phase.

An example is given in Fig. 7 for the 9th harmonic and phase 1, which even though it is not one of the highest impacting the distortion, from the data analysed it is the one which presents the highest variations in terms of angle range (at this harmonic order the accuracy is $\pm 9^\circ$). The range however, is not uniform and actually

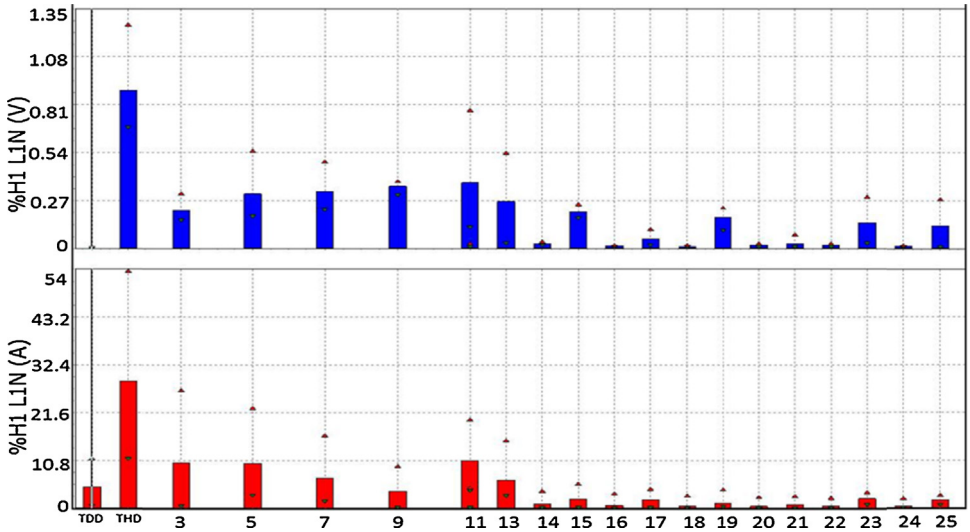

Fig. 6. Voltage and current harmonic histogram.

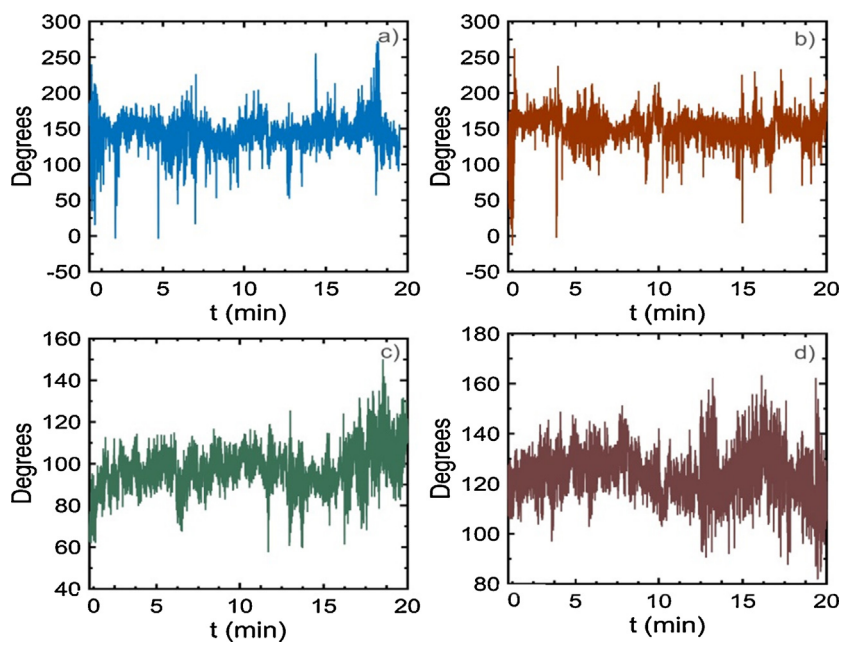

Fig. 7. Analysis of the 9th harmonic phase angle range (a) 1st, (b) 2nd, (c) 3rd and (d) 4th measurement for L_1.

tends to have a higher density (in terms of event number) around an average. This means that a statistical distribution can be drawn from all the harmonics and ranges.

Other harmonic frequencies are shown in Figs. A1–A6 in the Annex, and these also show preferential range angles.

Comparing the statistical distributions between the 4 measurements of all harmonics will provide the probability of two angles being within a range, i.e. if their amplitudes will tend to sum or subtract. Computing all events in Crystal-ball software, a distribution can be drawn with the corresponding mean value and standard deviation. This was performed for all the significant harmonics (odd harmonics until the 25th). Fig. 8 shows the statistical distribution of the 4 measurements when analysing the 9th harmonic variation shown in Fig. 7.

In this example the best fit distribution for the first and third measurements of the 9th harmonic correspond to a Logistic Distribution. Likewise the second and fourth measurements correspond to a Minimum Extreme Distribution. As can be observed from Fig. 8, even with higher angle variability, the mean values from their common referential (the fundamental) vary between 128° and 158° and from each other their difference will be below 90°. This means an amplitude summation is possible most of the time.

3.2. Comparison with the standard limits

As it was verified that the current and the THD_I changed during the charging cycle, for the most even analysis the TDD will be used, hence for the sake of the research analysis a higher focus will be given to the IEEE 519 standard, but it is also applicable to IEC standards. Apart from the TDD limit values presented in the standard, individual harmonics must also comply with the limits. From the harmonic histogram in Fig. 6, it can be verified that some harmonics such as the 11th could be out of the limits as set by both IEEE 519 and IEC 61000-3-12/2-4 standards. To identify the limits in Tables 1–4 the I_{SC} value was measured in the General Low Voltage Main Cabinet (PCC under analysis). Values ranged from 8080 A to 8480 A. The

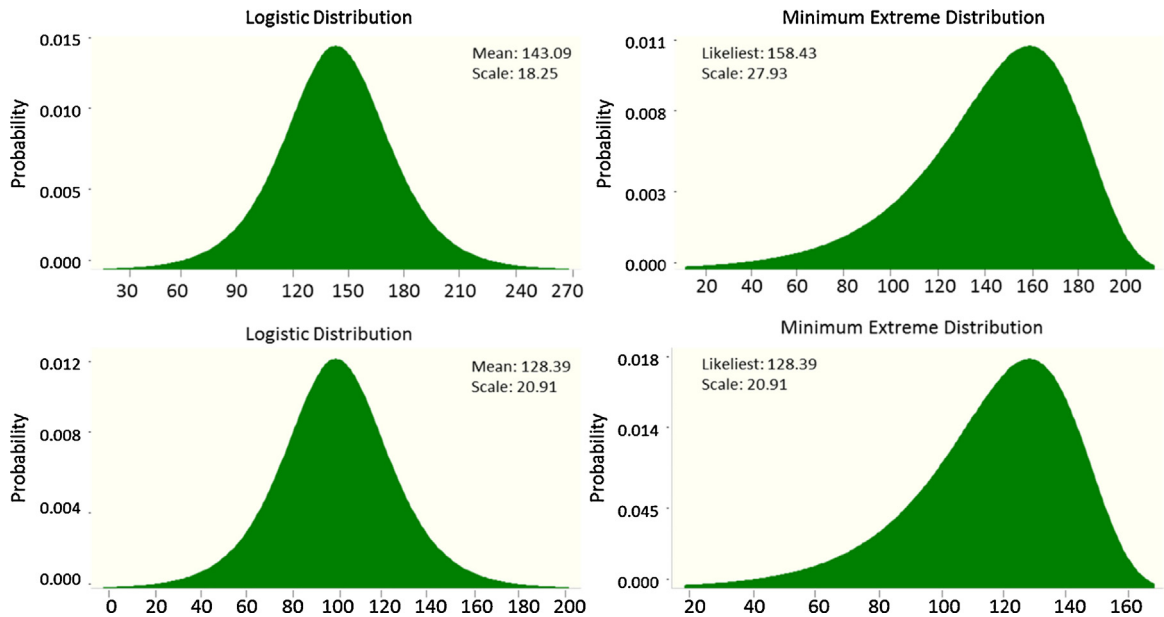

Fig. 8. Example of the 9th harmonic phase angles range statistical distributions for the four measurements.

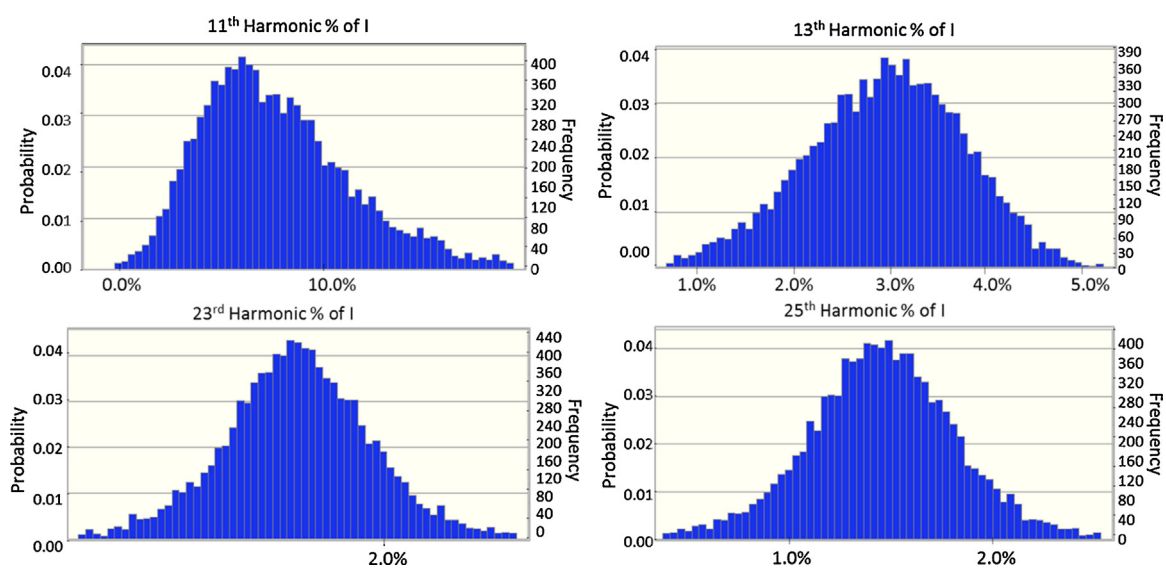

Fig. 9. Forecast of individual 11th, 13th, 23rd and 25th harmonics distortion.

I_{SC}/I_L was calculated considering the most unfavourable scenario: $I_{SC} = 8.08$ kA; $I_L = 67.5$ A → Ratio = 119.7

According to Tables 2 and 3, the TDD and THD limits are 15% and 16% respectively with the corresponding individual harmonic limits. Using the first measurement as an example, Crystal-ball tool was used to consider all values of the distortion during the cycle. The corresponding distributions regarding the individual impact of the harmonics were obtained, shown only for 11th, 13th, 23rd and 25th in Fig. 9. Is it possible that those harmonics may be in violation of the IEEE 519 and IEC 61000 3-12. Fig. 9 shows that the limits of 5.5% from Table 2 or 5% and 3% from Table 3 are exceeded by the 11th and 13th, and even though less probable, the 23rd and 25th may break the limit of 2% as well.

As seen in Fig. 4, TDD complies with the standard, but stays above 12% most of the charging cycle. It is important to point out that the ratio I_{SC}/I_L was close to, and just below 100. This would cause the limits of the TDD to be 12% and in this case the charger would fail to meet these requirements as well. Regarding the THD_V, Fig. 5 shows the voltage distortion being within the limit of 5%, and hence complying with the standard.

3.3. Simulating two electric vehicles

Simulating two vehicles will allow verifying if in fact there is no tendency for phase cancellation, meaning that TDD average value should suffer negligible change. Moreover, adding another fast charger/vehicle to the system will show how the harmonic limits of the standards to be consider change. To simulate two vehicles connected to the same feeder (both working at 67.5ARMS), two measurement cycles (1 and 2, Tables A1–A2 of the Annex) were considered. These two measurements represent two different vehicles charging, each with its own initial SOC, phase angles, and harmonic readings. All odd harmonics until 25th and corresponding data from a complete charging cycle were considered in Crystal-ball as inputs. Since the values of the phase angles are not always constant, the whole range was inputted as the best fit statistical distribution and

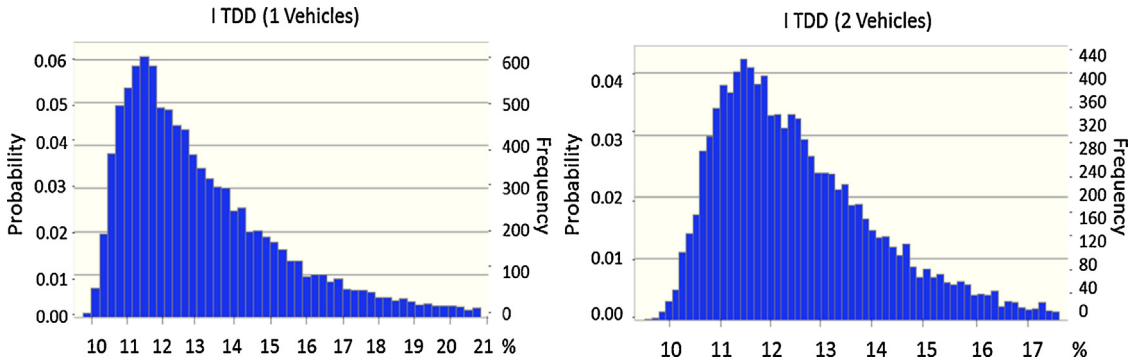

Fig. 10. TDD for 1 and 2 electric vehicle simulation.

ran for 10,000 trials. All other harmonics, not analysed individually, were considered and have a fix amplitude of 6.1 A (to reach the measured real TDD of 12%) and that their angles would add as well. The TDD was forecast for 1 and 2 vehicles and the results are shown in Fig. 10.

The simulation results with one and two vehicles report very close values with a mean of 12% for the TDD. As expected, with two vehicles the mean value slightly decreased (11.7%) since the amplitudes do not sum up perfectly. Furthermore the standard deviation seems to decrease when two vehicles are considered. This means that if the number of vehicles increase, the TDD will tend to be the same or to have a slightly decrease. Once again, this happens because preferential phase angles will make the amplitudes on average to add up ($\Delta\theta < 90$) but not perfectly (0°). Furthermore as the current I_L also increases linearly with the EV number, the distortion remains the same. However, if the I_{SC} is maintained this will eventually cause the violation of the limits set in the IEEE-519 standard as observed in Figs. 9 and 4. If more vehicles are connected this will not increasing the harmonic impact, however the I_L value will increase, making the ratio of I_{SC}/I_L decrease and causing the harmonic limits to be exceeded eventually (if infrastructure is unchanged). It is hence, extremely advisable that sufficiently high short-circuit power should be available at the interconnection point.

4. Conclusions

Three goals were set for this research, first to investigate the total voltage and current harmonic distortion impact caused by fast charging an electric vehicle and maintaining standard limit compliance. Second, understand how the total harmonic distortions caused by fast charger/EV load vary through the charging cycle. Finally, simulate if the distortions caused by charging more than one EV with the same charger model simultaneously, will decrease due to phase cancellation. THD_V, THD_I and TDD were calculated reporting the former at 1.2% and the latter two at 12% impacts respectively. For the charger under consideration, during the constant cycle stage, total values complied with the IEEE 519 and 61000-3-12/2-4 standard limits. However, individual harmonics failed to do so, mostly due to the 11th and 13th orders which are likely to exceed the 5.5% limit in IEEE 519 (5% and 3% respectively in IEC61000 2-4). Furthermore, also the 23rd and 25th harmonics even though less probable, may be in violation of their own individual limits. From the results of the experiment, it can be seen that the primary limitation of the number of chargers/vehicles in a cluster is not the power capacity of the upstream power transformers, but the harmonic limits for electricity pollution. For the standards limit analysis the TDD should be applied instead of the THD since there is a variation of the current during the cycle, and hence an opportunity for the European standard to improve in future versions. Regarding the resulting assessment with the simulation of two vehicles, it was verified that neither a synchronisation nor a random behaviour occurred in phase angles. Instead the phase angles tended to have preferential angle difference to the fundamental wave. From their statistical distribution study, one can observe that the differences between the same harmonic order (of each measurement) are lower than 90°, which means that with the same charger type they will add, suggesting that there is an upper limit to the number of vehicles to be considered allowable in the system. Should the number of EVs increase, i.e. I_L, the standard limits will decrease, reaching a point where harmonic limits are exceeded. It is therefore dependent on the robustness of the systems in terms of the foreseen short circuit current and the amount of current drawn by the vehicle cluster. Such types of infrastructure may need dedicated systems depending on the robustness (if short-circuit values are low) of the upstream system. Apart from the amplitude value of the harmonics, which may indeed differ between manufacturers (IGBTs, active filtering, etc.), the phase angle behaviour in other chargers should be verified.

Appendix A. Supplementary data

Supplementary data associated with this article can be found, in the online version, at http://dx.doi.org/10.1016/j.epsr.2015.05.012

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